Preliminary Sample-Size Analysis for CLAPS

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Abstract

CLAPS asks whether affectedness, how strongly an event affects the person or thing it happens to, influences the acceptability of passive sentences more than that of active ones, and whether that pattern holds across languages. The primary hypothesis (H1) holds that the affectedness slope is steeper for passives than for actives, and the secondary hypothesis (H2) that it is also steeper for passives than for pseudo-passives. This report evaluates by simulation what the planned design, 80 participants per language in at least 20 languages, can be expected to deliver. Both hypotheses are tested across languages in a two-stage synthesis: each language is analysed with the planned ordinal model, and the estimates are combined in a Bayesian random-effects meta-analysis. Which languages will take part is not yet known, so the simulations draw the effect of every language from a population whose mean and spread are estimated from the seven languages measured so far: English, Turkish and Norwegian in the CLAPS pilot, and Balinese, Hebrew, Indonesian and Mandarin in published studies. With 20 languages, the synthesis reaches a directional Bayes factor of 10 in 81% of simulated studies on H1, and in 82% on H2 across the 15 languages with a pseudo-passive. These rates allow for uncertainty about the size of the effect, and depend mainly on how much it varies between languages. They rise to 91% and 86.5% if the measured languages whose teams are already recruited take part, and at the estimated effect itself, the usual basis of a power analysis, they are 95% and 97.5%. In the pilots, the active interaction runs in the predicted direction in all three languages, and the pseudo-passive interaction in English but not in Turkish. Because affectedness is a property of the verb, the verbs bound the precision of each language's estimate.

Background

CLAPS (A Crosslinguistic Approach to Passive Semantics) asks whether the acceptability of passive and related sentence types tracks the semantic affectedness of the patient, whether that relationship differs by sentence type and whether it holds across languages. Affectedness, the degree to which a patient or theme undergoes a change of state or is causally impinged upon by the action, is a core dimension of argument structure (Beavers, 2011; Dowty, 1991). It has long been argued to be the semantic heart of the passive, which carries the meaning that the surface subject is in a state or circumstance characterised by the agent having acted upon it (Pinker et al., 1987). Acceptability-judgement and comprehension studies with adults in several languages support the view that passive acceptability is semantically structured (Ambridge et al., 2016; Ambridge et al., 2023; Aryawibawa & Ambridge, 2018; Bidgood et al., 2020; Darmasetiyawan & Ambridge, 2022; Liu & Ambridge, 2021).

The evidence to date spans English, Indonesian, Balinese, Hebrew and Mandarin. A Bayesian meta-analytic synthesis of those studies finds that affectedness raises the acceptability of passives but not of pseudo-passives (Ambridge et al., 2023). In most of the five languages, it raises passive acceptability more than active acceptability, although the size of that advantage varies considerably and in Balinese it runs the other way. The CLAPS pilot, run in English, Turkish and Norwegian, adds two languages to that evidence.

Each language's data will be collected by its own team, at a sample size fixed in advance. This report, prepared for the study's Stage 1 Registered Report, evaluates what the planned sample can be expected to deliver. It does so through a design analysis (Schönbrodt & Wagenmakers, 2018) whose simulated effects rest on the CLAPS pilot and on published estimates for four further languages.

The Model and the Predictions

Each acceptability judgement is a rating from 1 to 7. The ratings are modelled with a Bayesian cumulative-logit mixed-effects model, an ordinal model that respects the discreteness, ordering and bounded range of rating data (Bürkner & Vuorre, 2019). Treating the same ratings as continuous biases the regression coefficients and inflates error rates where floor or ceiling effects are present (Liddell & Kruschke, 2018), and modelling them cumulatively recovers psycholinguistic effects more accurately (Taylor et al., 2023; Veríssimo, 2021).

The model predicts acceptability from sentence type, affectedness and their interaction. Each stimulus appears as an active, a passive and a pseudo-passive. In English, for example, these are "The woman pushed the man", "The man was pushed by the woman" and "It was the man that the

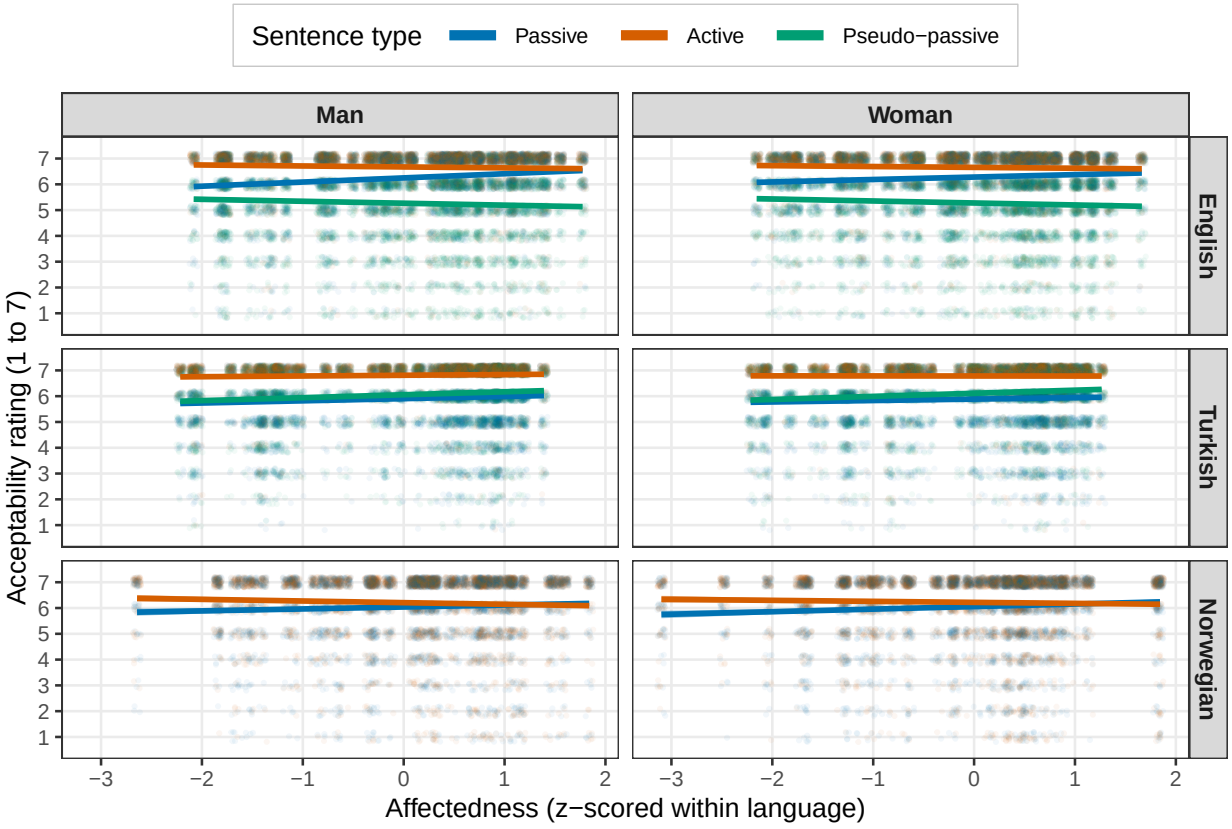
woman pushed", the last a cleft that puts the patient first without the passive's morphology. Each sentence type is treated as the same construction in every language that has it. The passive is the reference level, and the active and the pseudo-passive are compared with it.

H1, the primary hypothesis, holds that the affectedness slope is steeper for passives than for actives, so that the active-by-affectedness interaction is negative. H2, the secondary hypothesis, holds that the slope is also steeper for passives than for pseudo-passives, so that the pseudo-passive-by-affectedness interaction is negative. The passive slope itself carries no prediction. Affectedness might raise or lower the acceptability of sentences in general, and the theory requires only that its slope be steeper for passives, so each test compares the passive slope with another slope, whatever their signs. H1 is tested on the mean across all languages, and H2 on the mean across the languages with a pseudo-passive, which the Norwegian materials do not include.

The pilot, whose sample Appendix A describes, shows the predicted pattern in English and Norwegian (Figure 1). In both, the affectedness slope is positive and steepest for the passive, and flat or slightly negative for the active, as it is for the English pseudo-passive. In Turkish, the passive slope is also steeper than the active slope, though by less, while the pseudo-passive slope is at least as steep as the passive slope. The trends look much the same for male and female agents, although in English the active-by-affectedness interaction differs between them. The design therefore balances the two agent genders across participants (Appendix B).

Figure 1

Raw Pilot Acceptability by Affectedness, Sentence Type, Agent Gender and Language



Note. Panels are by language across rows and by agent gender across columns. Points are individual pilot ratings, jittered. Lines are linear trends for each sentence type, with 95% confidence bands that ignore the clustering of ratings within participants and verbs and so understate the uncertainty. Affectedness is the score normed for each agent-gender version, z-scored within language, and the model uses one score per verb, the mean of the two. The Norwegian materials include no pseudo-passive.

Design-Analysis Method

The design analysis simulates many studies and analyses each one as the real data will be analysed. Simulations of single languages, from the cumulative-logit model planned for each language's data, show how precisely one language's data estimate each interaction. Simulations of 20 languages then evaluate the confirmatory test, which combines the per-language estimates in the way described below. The procedure stays within the Bayesian framework throughout, because the confirmatory

test is a Bayes factor. Its result is therefore the probability that a study reaches a Bayes factor of 10, a different quantity from the power to return a significant p-value.

The per-language simulations take their effects from the pilot. For each language, the planned model is fitted to the pilot ratings. On every simulated study, the fixed effects and ordinal thresholds are drawn together from one draw of that fit's posterior, and the random-effect covariance is held at its posterior mean. New participants and new verb deviations are then drawn around the pilot's own verbs and their measured affectedness scores. Each reported rate is an assurance, a detection rate averaged over the effect sizes the pilot leaves plausible, which allows for the uncertainty of a single pilot's point estimate (Albers & Lakens, 2018). Every pilot language goes forward whatever its pilot showed, so the follow-up bias that Albers and Lakens describe does not arise.

Published effect sizes are subject to publication bias and tend to overstate true effects (Button et al., 2013; Gelman & Carlin, 2014), so design analyses anchored to them can be optimistic (Vasishth et al., 2018). In this report, published estimates enter only the multi-language synthesis, where they join the pilot estimates in estimating the population from which the simulated languages are drawn. Their average, −0.34 per standard deviation of affectedness, is close to the three pilots' −0.29, so they show no sign of inflation relative to the CLAPS protocol. Like the pilot data, the published data serve only to plan the study. Neither will enter the confirmatory tests, which will use only new data collected under the registered protocol.

Affectedness is a property of the verb. In the study, it will be rated in a separate norming task, in which twenty further speakers of each language rate the affectedness of the patient in each event, once for each agent-gender version. A verb's score is the mean of its two version scores, so each verb carries one affectedness score, the same across participants and sentence types. Every affectedness effect is therefore estimated from differences between verbs, and the verbs and the spread of their scores bound its precision, however many participants take part. This is the stimulus-sampling limit long recognised in psycholinguistics (Clark, 1973) and quantified for designs with crossed random effects by Westfall et al. (2014; see also Brysbaert & Stevens, 2018; Judd et al., 2017). With unlimited participants, the standard error of such an effect cannot fall below the by-verb standard deviation of that effect divided by the square root of the verbs' summed squared affectedness deviations. That floor can be read from the fitted pilot models without simulation.

The analysis model has the maximal random-effects structure the design supports (Barr et al., 2013), with correlated by-participant slopes for sentence type, affectedness and their interaction, and by-verb intercepts and sentence-type slopes. Because affectedness does not vary within a verb, a by-verb affectedness slope cannot be estimated and is omitted. Affectedness is centred and divided by twice its standard deviation within each language, and effects are reported per standard deviation of affectedness. The priors are centred at zero on the focal slopes, so they favour neither sign. The affectedness slope and the active interaction take normal priors with a standard deviation of 0.5 on the model's scale, or 0.25 per standard deviation of affectedness, and the pseudo-passive interaction

takes 0.6, or 0.30 per standard deviation. The other fixed effects take normal priors with a standard deviation of 1.5. The ordinal thresholds take Student-t priors with three degrees of freedom and a scale of 2.5, and the random-effect standard deviations Student-t priors with three degrees of freedom and a scale of 1. Their correlations take an LKJ prior with a shape of 2 (Gelman, 2006; Lewandowski et al., 2009).

In the confirmatory test, each language is first analysed on its own with this model, so it keeps its own rating thresholds, participants and verbs. A verb's effects are therefore estimated separately in each language. This suits verbs whose translations mean similar but never identical things, and across the meta-analysis languages verb effects correlate only about 0.2. The per-language posterior means and standard deviations of each interaction are then combined in a Bayesian random-effects meta-analysis, with language as the random effect. The mean across languages takes a normal prior with the standard deviation of the per-language prior, and the between-language standard deviation, τ, a half-normal prior with a scale of 0.5. One- and two-stage analyses of data from many studies give very similar results when each study is large and both make the same modelling assumptions (Burke et al., 2017), and each language here contributes 80 participants and about 72 verbs. A single model across all languages proved hard to fit (see Limitations), so it is kept as a planned robustness check, reported beside the confirmatory test but deciding neither hypothesis.

Evidence for each hypothesis is a directional Bayes factor, the posterior odds that the coefficient, or in the synthesis the mean across languages, carries the predicted sign divided by the same odds under the prior (Morey & Wagenmakers, 2014). The hypothesis is represented by the half of the zero-centred prior on the predicted side, a half-normal distribution of the kind *Nature Human Behaviour* names for representing a prediction. Its competitor, the reversed effect, is represented by the same half-normal on the other side. Under these symmetric priors, a Bayes factor of 10 corresponds to a posterior probability of .91 that the effect has the predicted sign, and a Bayes factor of 1/10 or less is equally strong evidence for the reversal. Within a language, the Bayes factor is estimated from posterior draws and from prior draws sampled alongside them. A simulated study counts as a detection when its Bayes factor reaches 10.

Each stimulus exists in two versions, one with a woman agent and one with a man agent. Each participant will see one version of each verb, so an English or Turkish session has 216 sentences, and the simulations describe that session. The versions are balanced across participants by counterbalanced lists (Appendix B).

The per-language simulations comprise 120 simulated studies at each of 70, 80 and 100 participants per language. The studies keep the pilot's verb counts, 72 in English and Turkish and 66 in Norwegian. Each study is fitted with four chains of 3,000 iterations at a target acceptance rate of .99 and a maximum treedepth of 12. After 1,000 warm-up iterations per chain, 8,000 post-warm-up draws remain. The confirmatory analysis will use sixteen chains of 5,000 iterations with 2,000 of

warm-up. These simulations serve to check the normal approximation with which the simulations of 20 languages represent each language's estimate (see Limitations).

The simulations ran on the University of Oxford's Advanced Research Computing service under R 4.4.2, with brms 2.21.0 and the cmdstanr backend on CmdStan 2.39.0, and the lockfile `renv.lock` fixes the package versions. The analysis code, the simulation settings and the aggregated results are openly available at https://github.com/pablobernabeu/CLAPS-analysis, where H1 and H2 carry the labels h1b and h2b. The registered analysis itself is in the repository's `stage2` folder, as numbered scripts with the order in which they will be run. This report rebuilds from those results without the raw pilot data, which are not yet public.

Results

The Pilot Estimates

Table 1 gives the two interactions in each pilot language, from the planned model fitted to the pilot ratings. The active interaction runs in the predicted direction in all three languages, most weakly in Turkish. The pseudo-passive interaction runs in the predicted direction in English but the other way in Turkish, where the affectedness slope is steeper for pseudo-passives than for passives. The simulations draw the Turkish effect from that posterior and so inherit the reversal. By contrast, the per-language models of the meta-analysis show the negative pseudo-passive interaction in Balinese, Indonesian and Mandarin.

Table 1

Pilot Estimates of the Two Interactions

Language	H1, active	H2, pseudo-passive
English	−0.44 [−0.67, −0.20]	−0.38 [−0.56, −0.20]
Turkish	−0.15 [−0.32, +0.03]	+0.16 [+0.03, +0.28]
Norwegian	−0.27 [−0.40, −0.14]	–

Note. Posterior means with 95% credible intervals, in logits per standard deviation of affectedness. Both interactions are predicted to be negative. The Norwegian materials include no pseudo-passive.

The Confirmatory Pooled Test

The collaborators expect at least 20 languages, 15 of them with a pseudo-passive, so the confirmatory test is simulated with 20 languages. Which languages will take part is not yet known, so the main

simulations do not assume that any of the seven measured so far, the three CLAPS pilots and the four other meta-analysis languages, is among them. Instead, every language's effect is drawn from a population whose mean and between-language spread are estimated from those seven. Each language's estimate is simulated at the precision that 80 participants and 72 verbs achieve, and each simulated study is analysed as registered. H2 is simulated in the same way in 15 languages with a pseudo-passive, drawn from the population estimated from the five measured languages that have one.

On these assumptions, the pooled test of H1 reaches a Bayes factor of 10 in 81% of 4,000 simulated studies (95% Monte Carlo interval [80, 82]), and the pooled test of H2 in 82% ([81, 83]). With 60 verbs in every language, the fewest the protocol allows, neither rate falls by as much as half a percentage point. Like the per-language rates, these are assurances, since each simulated study draws the population's mean and spread from their posterior. With the mean and spread held at their estimates (the posterior mean and median), as in a conventional power analysis, the rates are 95% [95, 96] for H1 and 97.5% [97, 98] for H2. Of the seven measured languages, six (all but Balinese) already have recruited teams. If they take part, each with its effect drawn around its estimate, the assurance for H1 is 91% [90, 92], and with the four of them that have a pseudo-passive, the assurance for H2 is 86.5% [85, 88].

A Bayes factor of 1/10 or less, strong evidence against a hypothesis, arises in 0.9% of simulated studies for H1 and 1.2% for H2, and in none at the estimated effect, so nearly every study that misses 10 is inconclusive. The assurances depend mainly on how much the effect varies between languages, which seven languages can estimate only roughly. More languages would add little, since no number of them makes up for a mean effect that proves close to zero: with 25 or 30 languages, H1 reaches 84%.

The Sampling Plan

Every team will collect usable data from 80 participants, replacing any participant who is excluded. The number is fixed in advance because each team must state its sample size to its ethics committee. Each language will use as close to 72 verbs as its supply of suitable verbs allows. Every language enters the pooled tests whatever its pilot showed, Turkish included, since leaving a language out on the strength of its pilot would amount to choosing languages by their results. The H2 rates above already allow for the Turkish reversal, since Turkish is one of the five languages behind its population. Table 2 summarises the planned analyses and their detection rates.

Table 2

Planned Analyses and Their Detection Rates

Analysis	Coefficient and direction	Test	Detection rate
H1, primary	Active by affectedness, mean across languages, negative	Directional Bayes factor of 10 in the two-stage synthesis	95% at the estimated effect, 81% to 91% allowing for its uncertainty
H2, secondary	Pseudo-passive by affectedness, mean across languages, negative	As for H1, in the languages with a pseudo-passive	97.5% at the estimated effect, 82% to 86.5% allowing for its uncertainty
Robustness check	Both interactions in a single model across languages	Reported with its directional Bayes factors, deciding neither hypothesis	Not used to plan the sample

Note. The rates are at 80 participants per language, with 20 languages for H1 and 15 for H2. Allowing for uncertainty about the effect, the lower rate draws every language from the population estimated from the measured languages, and the higher lets the measured languages with recruited teams take part.

Nature Human Behaviour sets its 0.95 power requirement for frequentist analysis plans. For a Bayesian plan, it asks that data collection continue until the Bayes factor is "at least 10 times in favour of the experimental hypothesis over the null hypothesis (or vice versa)". Authors with limited resources may instead state a maximum feasible sample, "sufficiently large that inconclusive results at this sample size would nevertheless be an important message for the field". CLAPS takes the second route for both hypotheses. Each team stops at 80 participants, the most the collecting teams can provide, and data collection closes six months after the minimum of 20 languages is reached, whatever the Bayes factors. A test whose Bayes factor has reached neither 10 nor 1/10 by then is reported as inconclusive. At that sample, each language is close to the limit its verbs set, so further participants would add little. An inconclusive result from at least 20 languages and 1,600 participants would still be an important message for the field.

Limitations

At 80 participants, no simulated per-language fit has a divergent transition or a saturated treedepth. Confirmatory fits must also reach an R-hat below 1.01 and bulk and tail effective sample sizes of at least 400. With the shorter runs used for simulation, 27 of 120 English fits meet those thresholds,

as do 33 Norwegian and 12 Turkish fits. The fits that met them gave the same results as the others, within what random subsets of their size would show. With longer chains and four times as many of them, the confirmatory fits will have six times the post-warm-up draws.

Fitted to the combined pilot data, 75,528 ratings over 210 verbs, a single model across languages estimated the interaction at about −0.24 logits per standard deviation of affectedness, with a 95% credible interval from −0.51 to +0.15. That is close to the average of the three pilot estimates, −0.29, but the model is much harder to fit than the per-language ones. It saturated the maximum treedepth on three quarters of its iterations, a problem longer chains do not fix. In simulations with 20 languages of 20 verbs each, no form more complex than one with sentence-type slopes for participants and verbs and uncorrelated by-language slopes converged. The robustness check's model ladder is therefore expected to end at that form.

The 20-language figures rest on an assumed population of languages. The seven measured so far are a convenience sample from five families in two macroareas, and the simulations treat every language as a draw from a population like them. They also give each language's estimate the standard error of a normal approximation to its analysis. The calibration behind the rates for a language without a pilot widens that standard error by a factor of 1.17, and with it neither of the rates with no measured language taking part falls by more than 0.4 of a percentage point.

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Appendix A

Composition of the Pilot Sample

This appendix describes the pilot sample whose ratings give the descriptive trends in Figure 1 and to which the planned model is fitted for the simulations. Table A1 gives the counts, read directly from the harmonised pilot data.

Table A1

Composition of the Harmonised Pilot Sample, by Language

Language	Participants	Verbs	Sentence types	Items	Judgements
English	70	72	3	144	30,240
Turkish	70	72	3	144	30,240
Norwegian	57	66	2	132	15,048

Note. Items are verb-by-agent-gender combinations, so the item count is twice the verb count. Every pilot participant saw both versions of every verb, whereas in the study itself each participant will see one version of each verb (Appendix B). The Norwegian materials include no pseudo-passive, hence only two sentence types. Norwegian sessions also included a synthetic passive, which the preregistered analysis excludes, so a Norwegian participant rated one and a half times as many sentences as this table counts.

Appendix B

The Agent-Gender Lists

This appendix sets out how the two agent-gender versions of each stimulus are balanced across participants, and why. Each participant will see one version of each verb, so the balance has to come from the assignment of versions to participants.

The versions need balancing because agent gender moves the interaction in English. A linear mixed model of the 1-to-7 pilot ratings, used here only to gauge the relative size of the gender term, codes agent gender as plus or minus a half. In it, the English interaction is -0.16 rating points per standard deviation of affectedness, and its gender-by-active-by-affectedness term is $+0.068$ ($SE = 0.023$). The interaction is therefore markedly stronger for man-agent items than for woman-agent ones. A design in which every participant saw only woman-agent versions, or only man-agent ones, would move the English interaction by about 21%, against 1% in Turkish and 3% in Norwegian.

Counterbalanced lists remove that shift without lengthening the session. Each list shows woman agents on half the verbs and man agents on the other half, with the same agent across a verb's sentence types, and across the lists each verb appears equally often with each agent. Each participant therefore sees both agent genders equally often, each verb is rated in each version by half the sample, and the estimand remains the slope averaged over agent gender. The protocol uses eight such lists, although two would be enough. With two, the verbs would be split into two sets of 36, or 33 in Norwegian, matched on affectedness and verb class. One list would show the first set with woman agents and the second with man agents, and the other the reverse. As long as every list shows both agent genders on matched verb sets, unequal numbers of participants across the lists leave the interaction essentially unchanged. Even so, assigning participants to the lists in turn, and replacing exclusions within the same list, keeps each verb's two versions evenly rated. Splitting the agent-gender versions across participants in this way follows Ambridge et al. (2016, Study 3) and the Hebrew study of Ambridge et al. (2023).

Both agent-gender versions should stay in the affectedness rating task, which asks for one judgement per verb and agent gender and is short. Halving it would leave each agent-specific score resting on half as many raters and would attenuate every slope estimated from those scores.